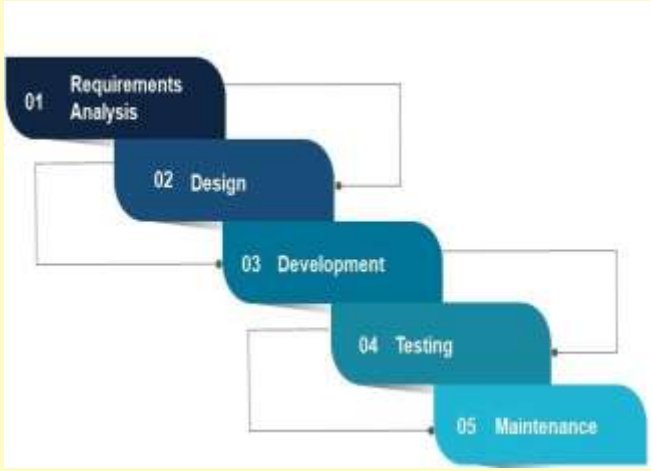


CSE 470 – Agile Methodology

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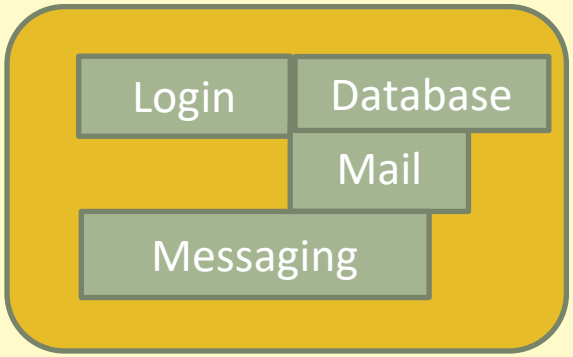
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Why we need Agile



1. Now a days, the requirement of software changes frequently, even it can change a number of times in a day.

2. For live systems, like Amazon, Facebook a few seconds of downtime can cause a lot. Changes in such cases needs to be smooth enough so that customers interaction do not interrupt.

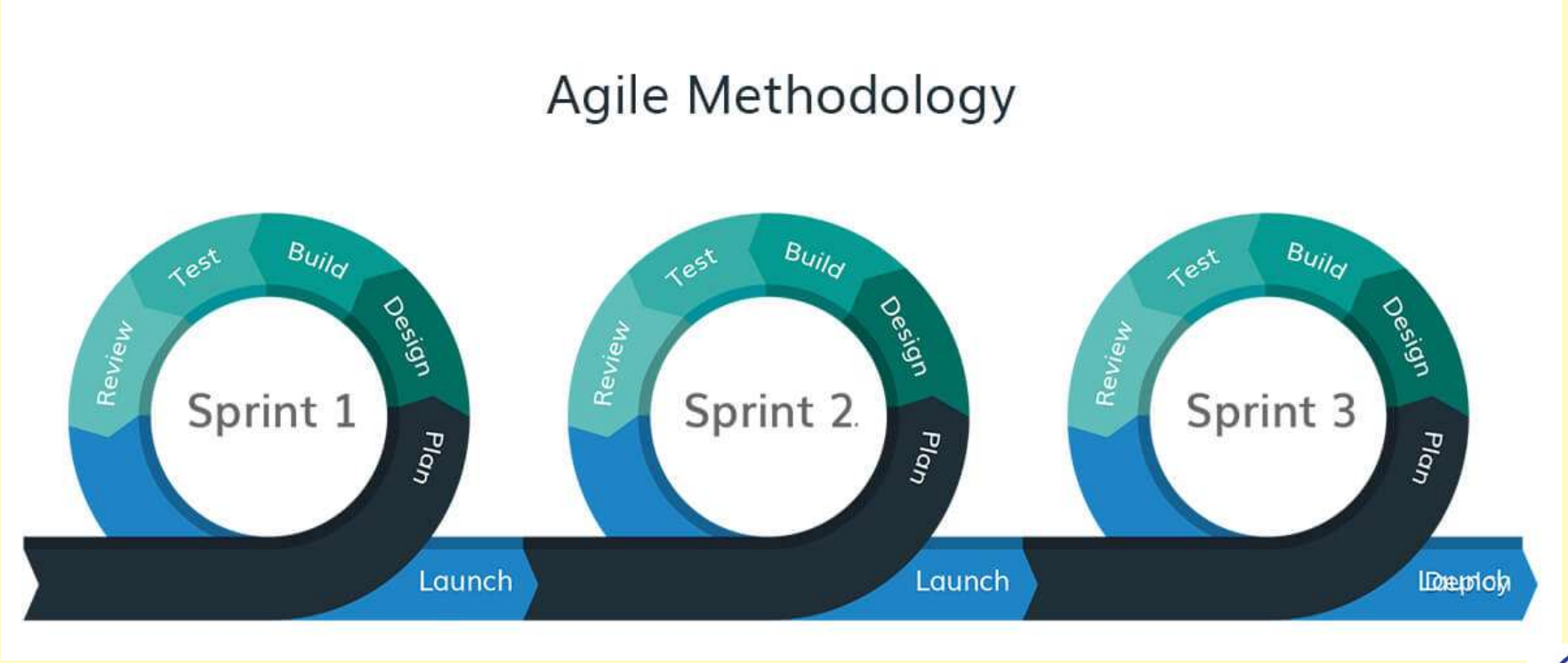


Read to know more about Monolithic architecture: click here
Read to know more about Agile benefits: click here

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What is ‘Agility?’

- *Agile is a set of principles and values. It is a practice to be followed.*
- *Agile is a combination of Iterative and Incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software.*



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Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it.





Individuals and interactions	*over*	***processes and tools***
Working software	*over*	***comprehensive documentation***
Customer collaboration	*over*	***contract negotiation***
Responding to change	*over*	***following a plan***

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Agile Methodologies

- Extreme Programming
- Agile Unified Process (Excluded for Summer 25)
- Scrum

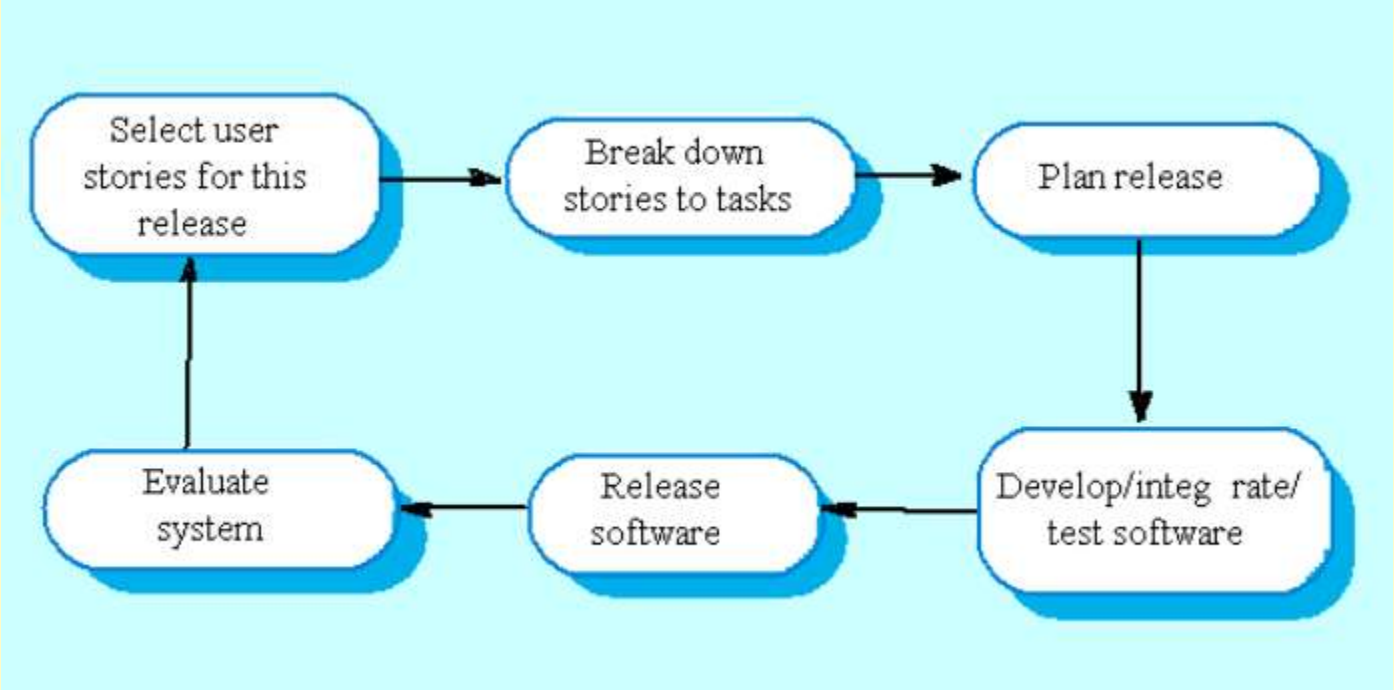
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Extreme Programming

- In short also known as **XP**
- New versions may be built several times per day
- Increments are delivered to customers every 2 weeks
- All tests must be run for every build and the build is only accepted if tests run successfully.

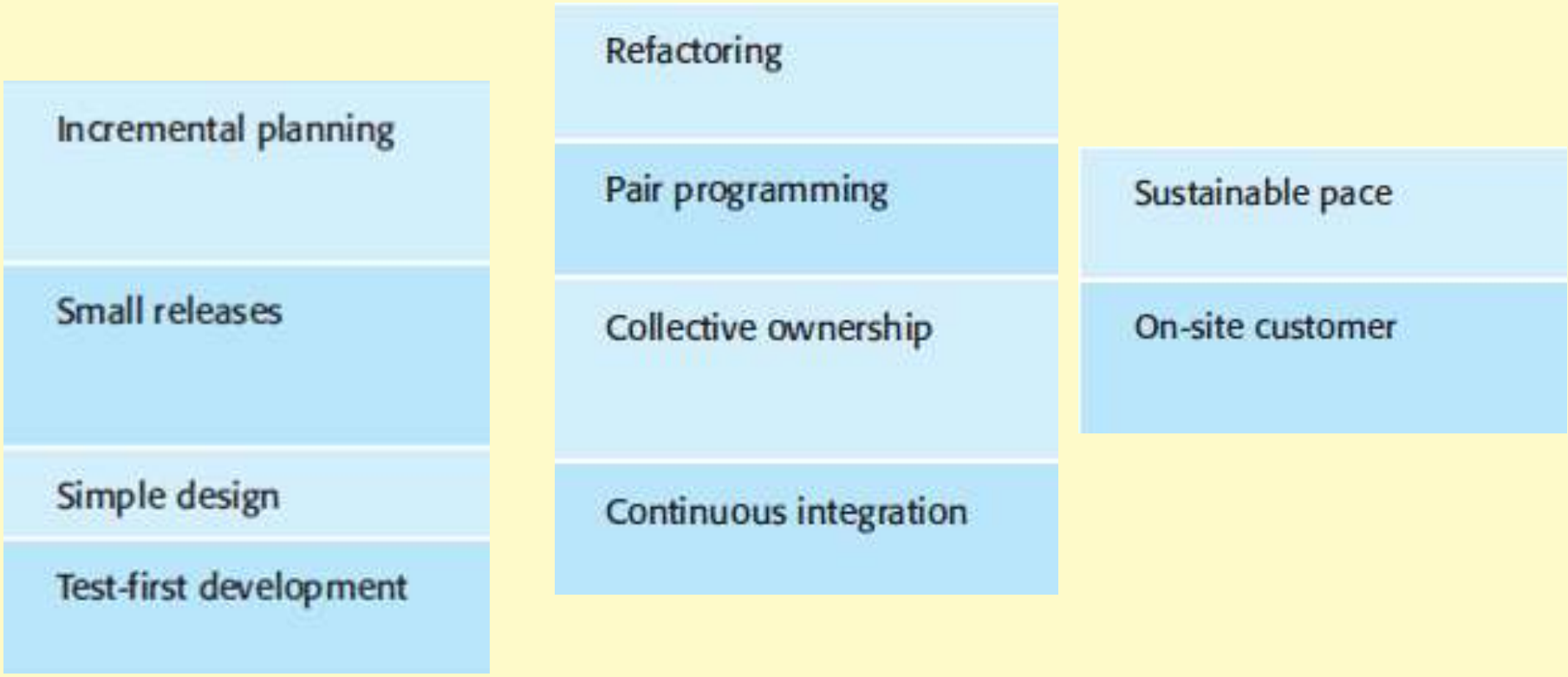
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XP Workflow



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XP Principle or Practice



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Pair Programming

- Two people code in share, they switch roles from observer to -navigator frequently
- Observing code produces better code, so less business cost.
- Support each other, better chance for learning.



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Unit Testing

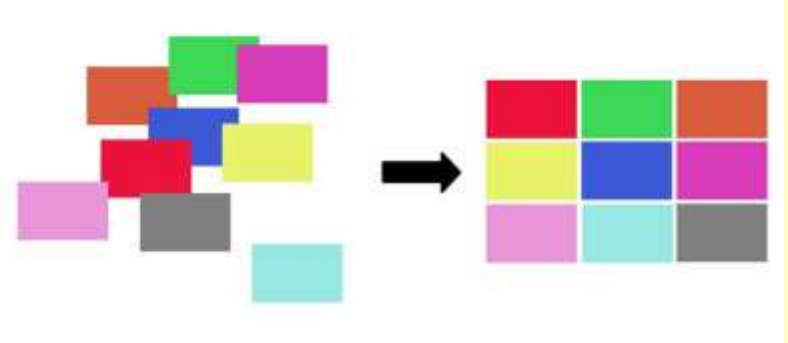
- Writes unit test for each method to identify problems and resolve it
- Test First Development (TFD) is used to ensure better code quality and less error. It refers to write test code first and after that write the development code.

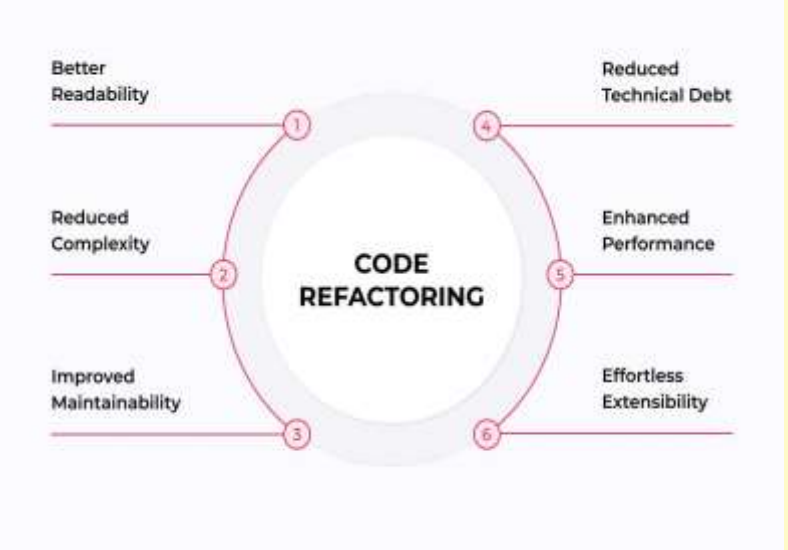


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Refactoring

- It’s a technique for restructuring existing code by changing its internal structure without changing its external behaviour.
- Refactoring includes reducing duplicate codes, breaking long classes/methods into small ones, appropriate variable naming and many more.
- It is done for requirement change, improving design and easy extensibility of software.





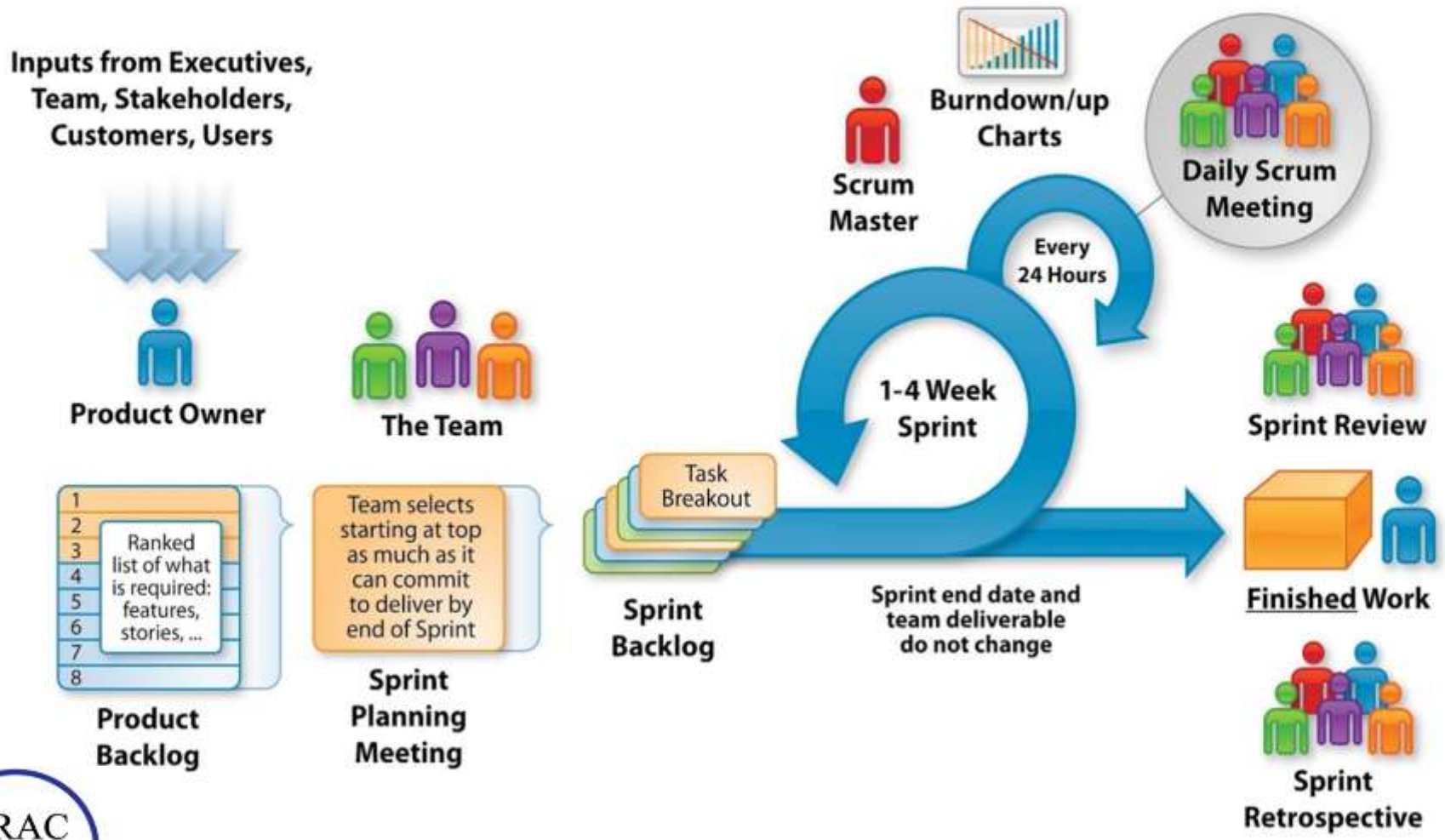
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Scrum

- It is the most widely used Agile S/W development method for project management
- The full software is delivered in 1-4 weeks iterations

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The Agile - Scrum Framework



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Scrum Framework

Roles

- Product owner
- Scrum Master
- Team

Ceremonies

- Sprint planning
- Sprint review and Sprint retrospective
- Daily scrum meeting

Artifacts

- Product backlog
- Sprint backlog
- Burndown charts

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Scrum Roles



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Scrum Roles

– Product Owner
  - Possibly a Product Manager or Project Sponsor
  - Decides features, release date, prioritization, $$$
  - Conveys a clear vision to the Scrum team
  - Replies to frequently asked queries



– Scrum Master
  - Typically a Project Manager or Team Leader
  - Responsible for enacting Scrum values and practices
  - Remove impediments / politics, keeps everyone productive

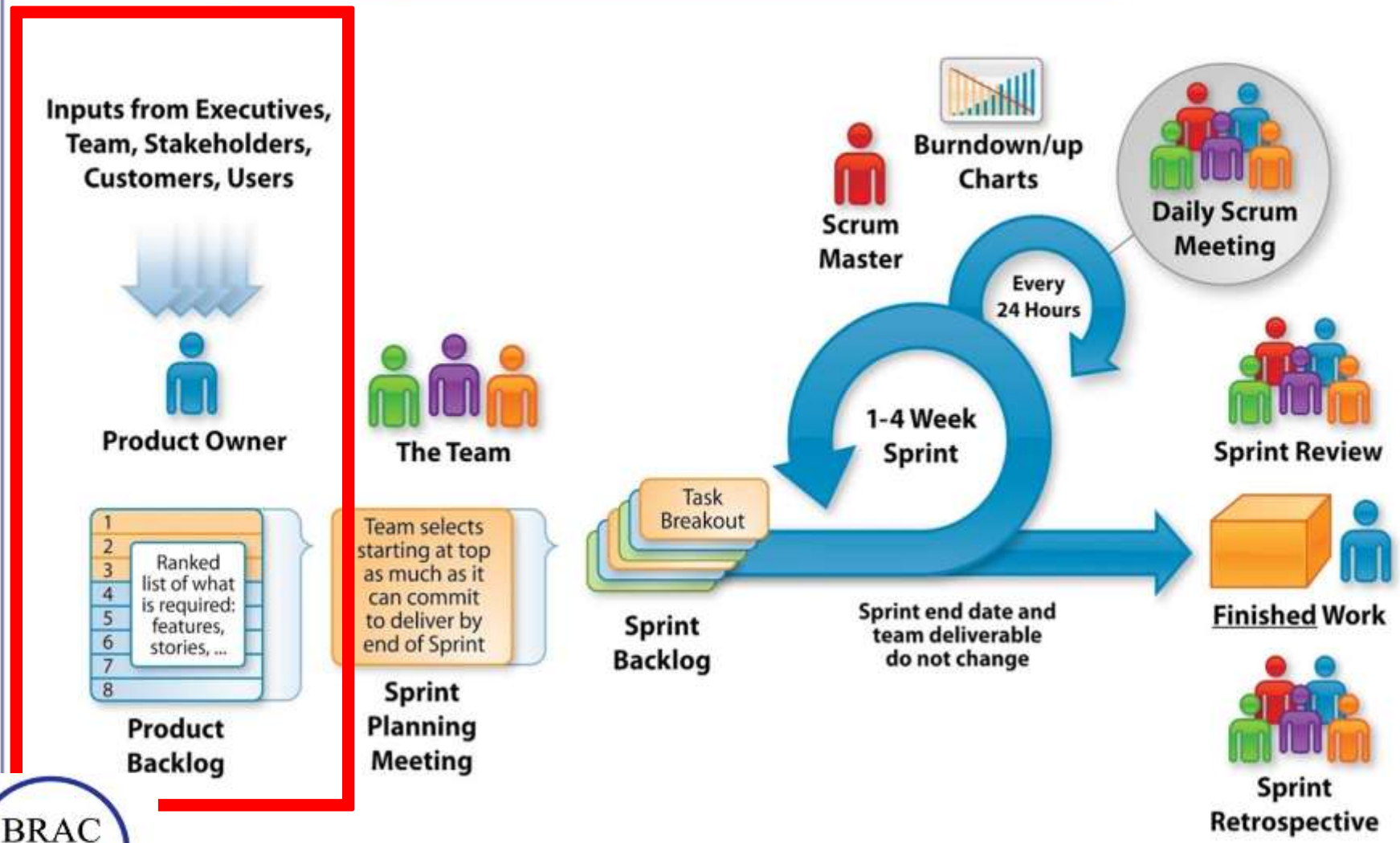


– Project Team
  - 5-10 members; Teams are self-organizing
  - Cross-functional: QA, Programmers, UI Designers, etc.
  - Membership should change only between sprints



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The Agile - Scrum Framework



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Artifact - Product Backlog



This is the product backlog

- The requirements
- A list of all desired work on project
- Ideally expressed as a list of user stories along with "story points", such that each item has value to users or customers of the product
- Prioritized by the product owner
- Reprioritized at start of each sprint

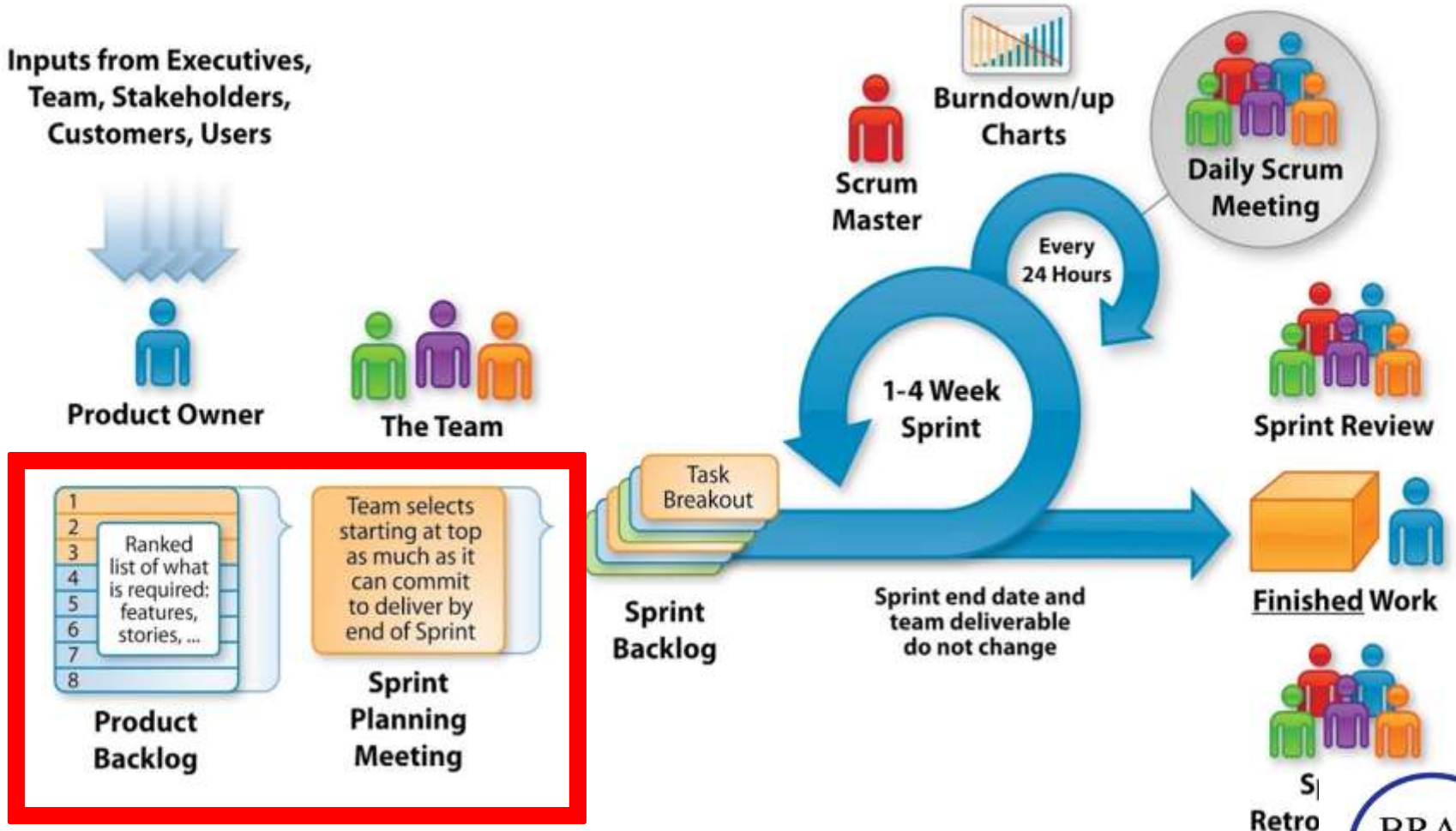
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Artifact - Product Backlog

Feature	Backlog item	Estimate
A	Allow a guest to make a reservation	3 (story points)
B	As a guest, I want to cancel a reservation.	5
C	As a guest, I want to change the dates of a reservation.	3
D	As a hotel employee, I can run RevPAR reports (revenue-per-available-room)	8
E	Improve exception handling	8
…	…	30
…	…	50

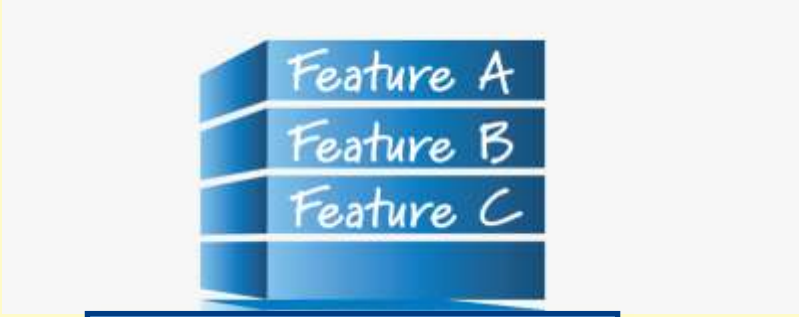
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The Agile - Scrum Framework



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Artifact - Sprint Backlog



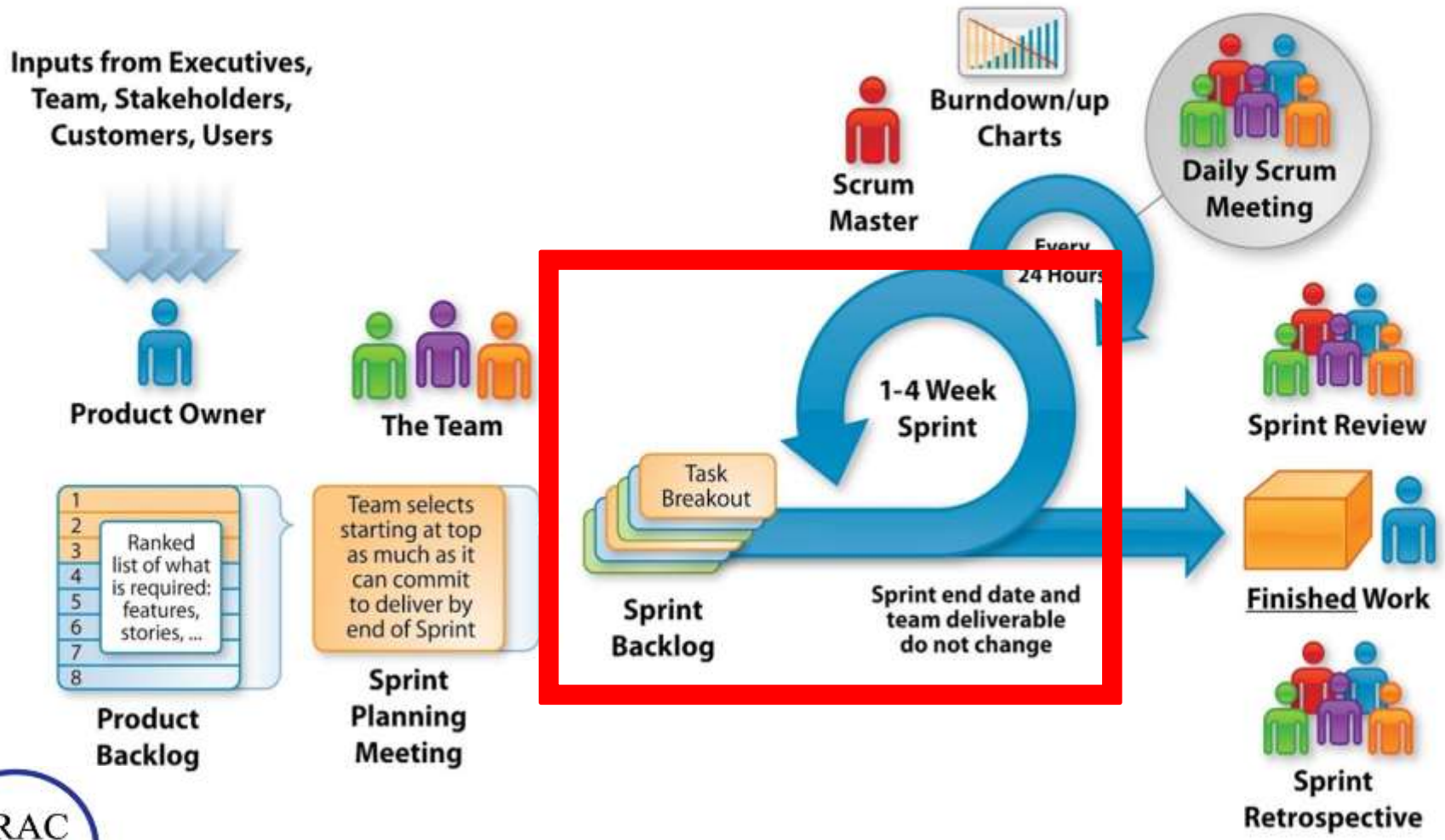
This is the sprint backlog

- High priority features are selected based on points from product backlog.
- These features will be covered within that 1-4 weeks time slot called as **sprint**.
- Estimated work remaining is updated daily
- Any team member can not add, delete change sprint backlog

If the work turns out to be different than they expected, they collaborate with the Product Owner to negotiate the scope of the Sprint Backlog within the Sprint without affecting the Sprint Goal.

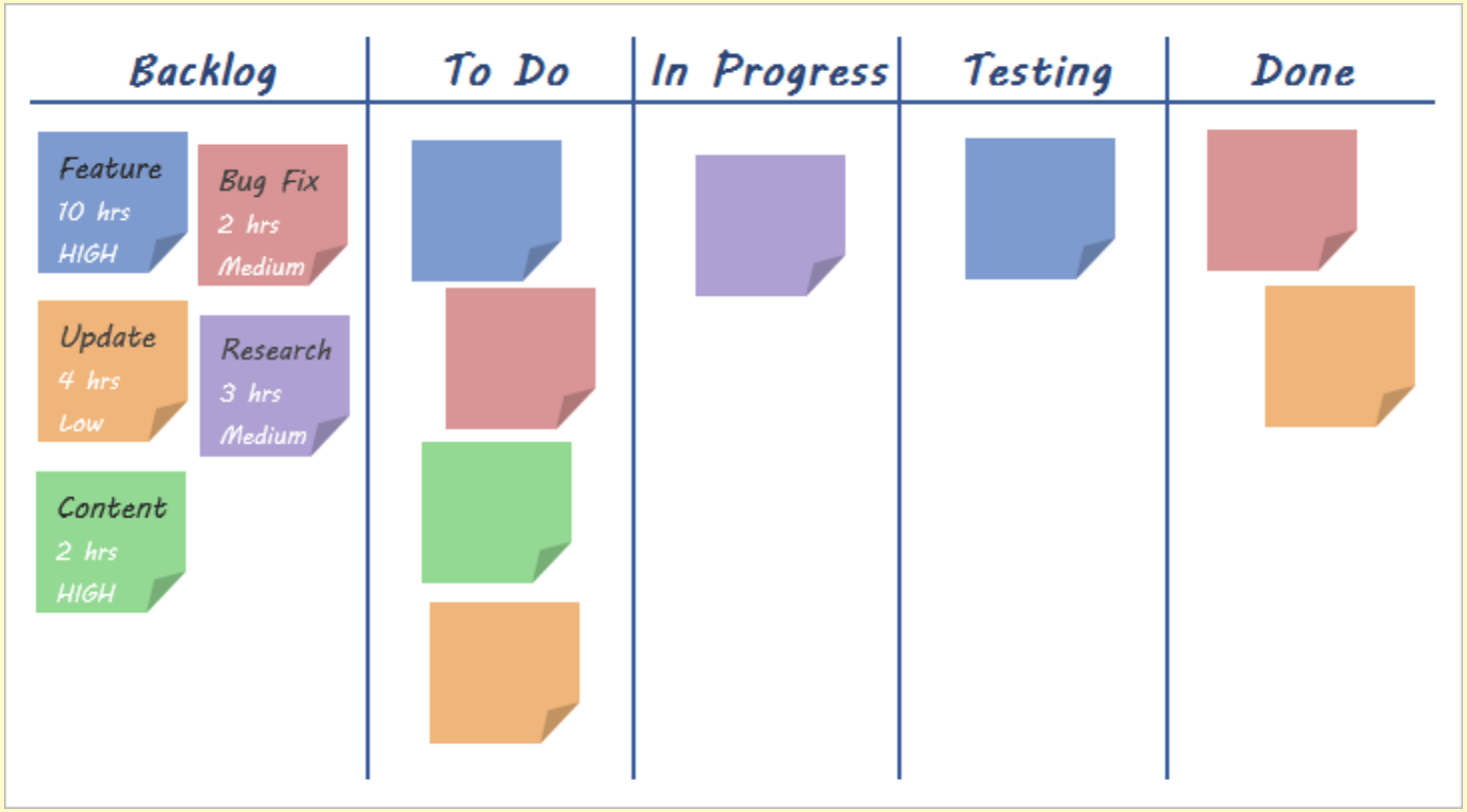
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The Agile - Scrum Framework



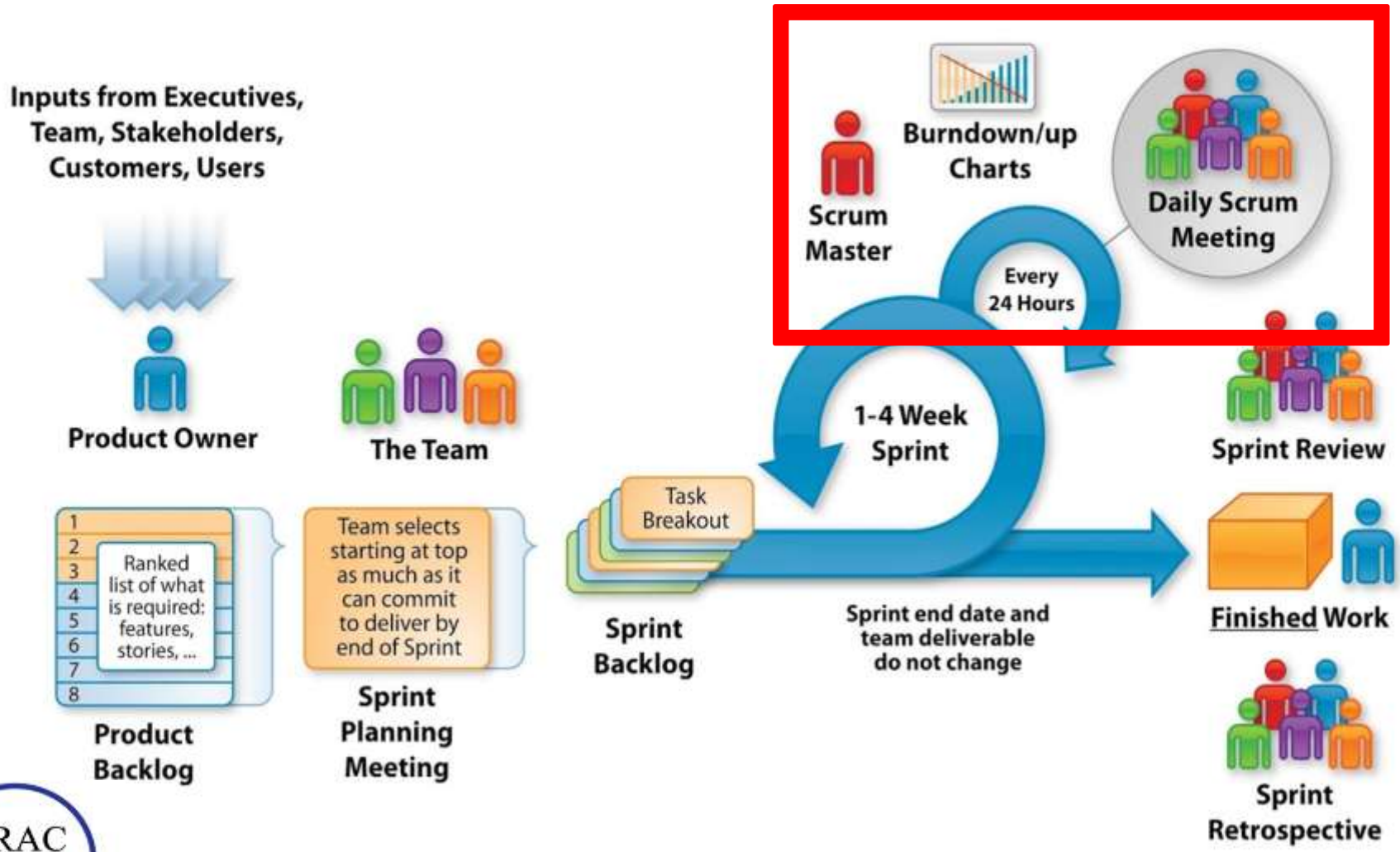
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Artifact - Sprint Board



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The Agile - Scrum Framework



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Ceremony - Daily Scrum Meeting



- Parameters
  - Daily, ~15 minutes, Stand-up
  - Anyone late pays a $1 fee
- Not for problem solving
  - Whole world is invited
  - Only team members, Scrum Master, product owner, can talk
  - Helps avoid other unnecessary meetings
- Three questions answered by each team member:
  1. What did you do yesterday?
  2. What will you do today?
  3. What obstacles are in your way?

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Artifact – Burndown Chart

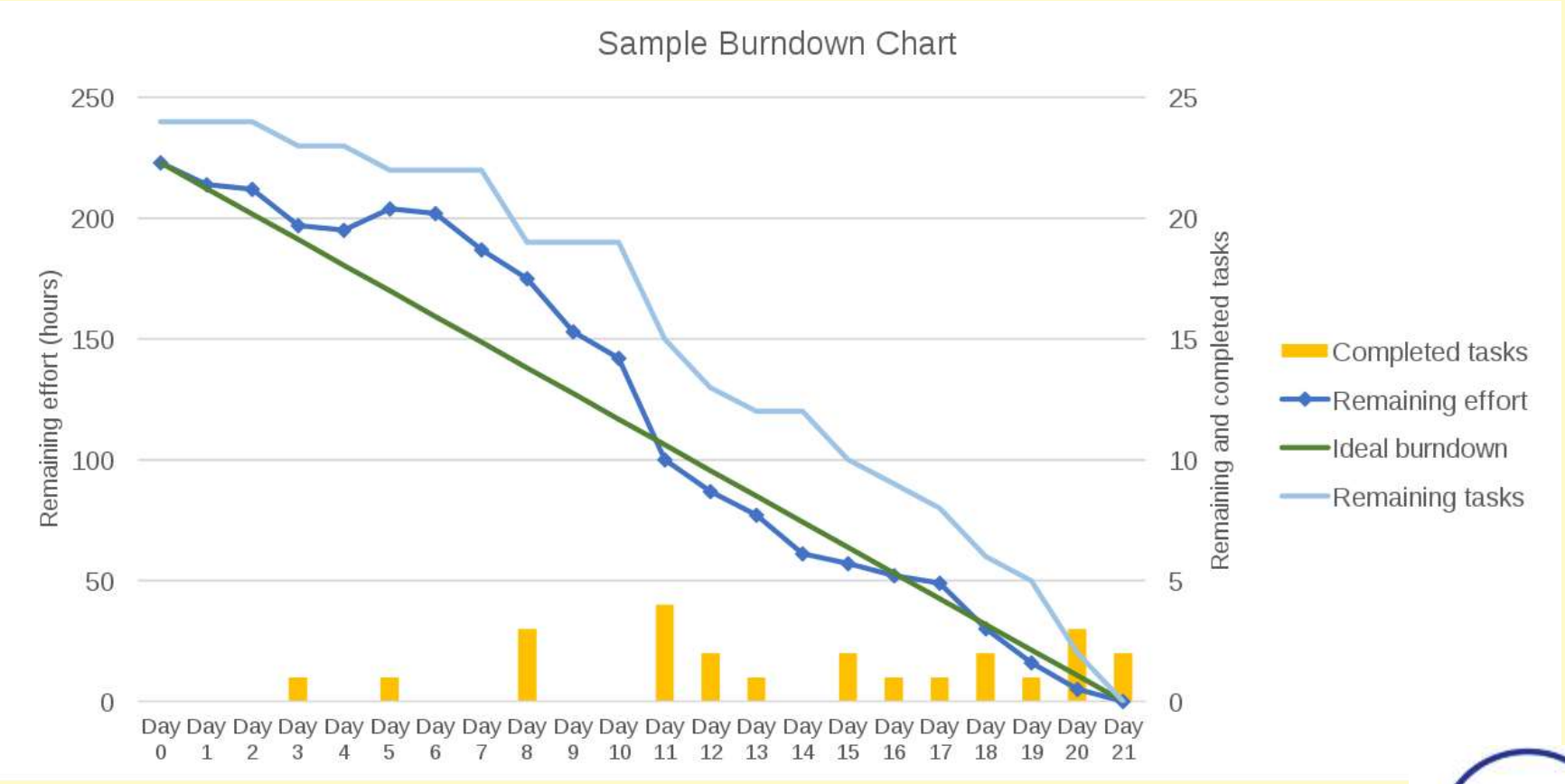
- A display of what work has been completed and what is left to complete
  - one for each developer or work item
  - updated every day
  - (make best guess about hours/points completed each day)

- *variation:* Release burndown chart
  - shows overall progress
  - updated at end of each sprint

2 min read: https://www.mountaingoatsoftware.com/agile/scrum/scrum-tools/release-burndown

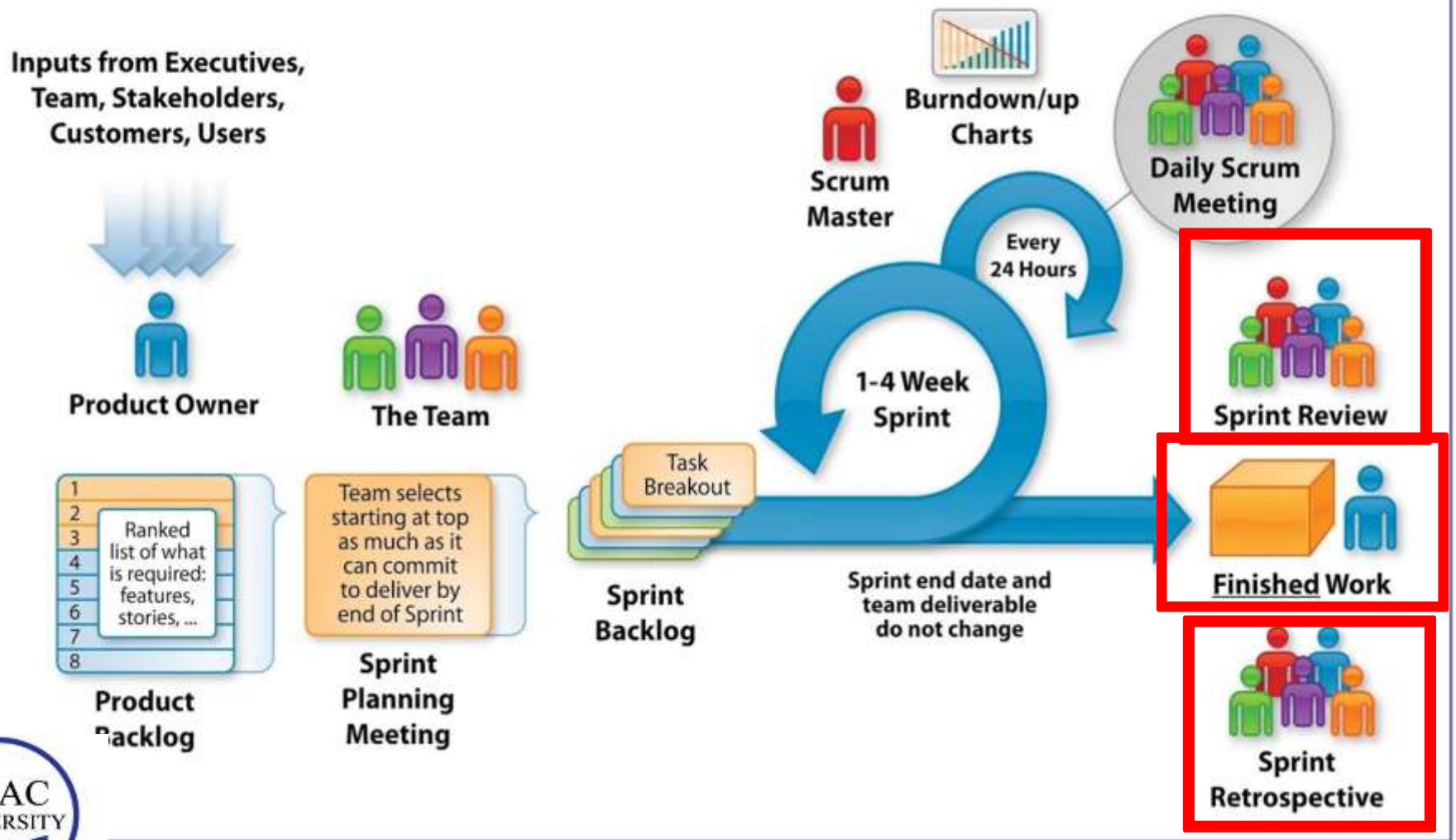
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Artifact – Burndown Chart



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The Agile - Scrum Framework



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Ceremony - Sprint Review

- Team presents what it accomplished during the sprint
- Typically takes the form of a demo of new features or underlying architecture
- Informal
  - 2-hour prep time rule
  - No slides
- Whole team participates
- Invite the world



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Ceremony – Sprint Retrospective

- Identify the scopes of improvement for better result in next sprints
- What worked well, what went wrong are also discussed



2 min read: https://www.scrum.org/resources/what-is-a-sprint-retrospective

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Summary

- We have learnt three types of agile principles used in the industry.



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